

BukuWarung: Modernizing accounting for Indonesian MSMEs in the cloud

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ABOUT BukuWarung

With Google Cloud, BukuWarung is redefining the ambition of digitizing accounting and cloud processes for up to 60 million MSMEs in Indonesia.



About BukuWarung

BukuWarung aims to build a digital infrastructure for 60 million micro, small and medium enterprises across Indonesia, based initially on a technology platform that provides bookkeeping and digital payment products. Founded in 2019, the business has grown to 300 employees.

**Google
Cloud
results**

**Builds
financially
literate**

Industries: Financial Services &
Insurance

- Scales seamlessly to ingest 70+ data points per merchant for every 100 transactions
- Enables data analytics team to support up to 500 team members without adding headcount
- Frees analytics team to spend 30% more time building better reports and obtaining deeper insights
- Controls budgets more effectively by keeping

challenges
member
extension
themes
development
product
improvement
experience

Location: Indonesia

(<https://workspace.google.com/products/gmail/>)



About Searce

Searce is a niche cloud consulting business that specializes in cloud data engineering, AI/ML, and advanced cloud infrastructure services. The business is one of Google Cloud's top partners, with 3,000+ clients moved successfully to the cloud.

Google Cloud (<https://cloud.google.com/>)

business that specializes in cloud data

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Dataprep

(<https://cloud.google.com/dataprep/>)

Firebase (<https://firebase.google.com/>)

3,000+ clients moved successfully to the cloud.

Looker (<https://cloud.google.com/looker/>)

Google Workspace

(<https://workspace.google.com/>)

Gmail

(<https://workspace.google.com/products/gmail/>)

**unexpected
cost spikes
below 15%**

Micro, small and medium enterprises (MSMEs) are the lifeblood of communities and commerce in Indonesia, contributing nearly two-thirds of the country's gross domestic product. Of the 60 million MSMEs in the country, 3.5 million are warungs, small, family-owned stalls that service communities across the nation's 17,000 islands. These traditional stalls typically tracked orders, sales, and other administrative tasks using pen and paper, making financial tracking difficult and hampering attempts to obtain credit.

BukuWarung (<https://bukuwarung.com/>)'s founders aim to bring these MSMEs into the digital age, replacing hastily-scrawled notes with financial products offered over a cloud-based platform. Established in 2019, the business released an accounting product that attracted one million users within six months, and has now grown to 300+ employees across Indonesia, India, Singapore, and Vietnam. BukuWarung then launched a service that enabled MSMEs to send and receive money.

To build products relevant to MSMEs in Indonesia and beyond, BukuWarung needed to analyze burgeoning volumes of data. In addition, during its startup phase, the business had to track application usage closely and enable potential investors to monitor its growth. Google Cloud provided the scalable, proven cloud services that could meet

these requirements and enable BukuWarung to realize its vision.

Building an all-in-one financial platform for MSMEs

With a vision to build an all-in-one financial platform for the MSME community, the business's offerings now include digital payments, an in-application wallet, and buy now, pay later services. It also offers cash loans to merchants based on electronic payments data records captured, and business management software and digital bookkeeping products.

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*—Alka Gupta, Director of
Data, Marketing, Business
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BukuWarung knew harnessing data would be key to its longer-term success. As a hungry startup growing 20% week-to-week on occasion, the business had worked with a database that an engineer had to update manually each morning so BukuWarung team members could run queries and generate reports. However, as the business matured and moved toward compliance with Indonesia's financial regulations, its leadership began exploring technology platforms that could support growth and regulatory requirements.

The company turned to the BigQuery (<https://cloud.google.com/bigquery>) data warehouse with Looker (<https://www.looker.com/>) analytics and visualization, and the Firebase (<https://firebase.google.com/>) application development platform, to meet its needs. "Our plan was to adopt a scalable, single source of truth, indisputable across

everything we did, every partnership we had and every service we worked with," explains Alka Gupta, Director of Data, Marketing, Business Operations & People, BukuWarung. "We also wanted a solution that was effectively 'plug and play,' fast and easy to use. BigQuery was ideal for our needs."

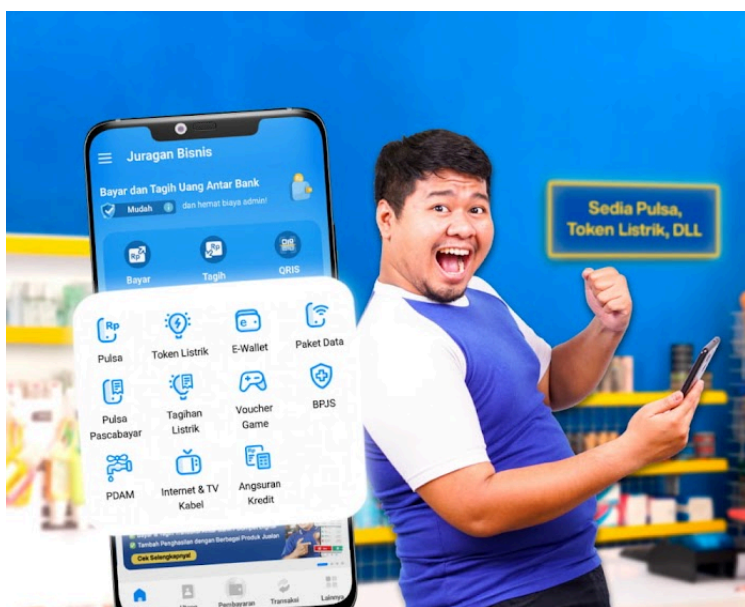
The business then worked closely with the Google Cloud team to deploy BigQuery, Firebase and Looker across its business. It also harnessed the support of Searce (<https://www.searce.com/>), which provided expert licensing advice and support across these products, and Google Workspace (https://workspace.google.com/intl/en_sg/) for day-to-day operations.

Standardizing data at scale for transparency to gain deeper insights

With BigQuery, BukuWarung ingests insightful data points and processes these into reliable datasets used for reporting and downstream analysis. The business has also created data marts that reside on top of BigQuery tables to meet individual departments' specific needs. The data warehouse is meeting BukuWarung's scalability requirements. Every merchant acquired by the business provides about 80 data points in the first week on board, and about 70 data points for every 100 transactions thereafter. The business has grown to several million customers and aims to reach as many of the 60 million MSMEs in Indonesia as possible.

With BigQuery and Looker, BukuWarung has standardized the data used by analysts, data scientists, and business users to collaborate and perform their roles, positioning the business to pursue certification to International Organization for Standardization (ISO) quality standards, build presence and credibility within the fintech market, and work with banks. The business complies easily with Indonesia's strict data protection regime, as BigQuery's audit and authority features are rigorous enough to meet the needs of regulators and banks.

Looker now provides the visualization and dashboards that enable the business to view the reports that show how key internal metrics and overall performance is tracking. The seamless integration between Google Cloud infrastructure and data services and Google Workspace means the business can automate the distribution of up-to-date reports to relevant team members over Gmail.



Delivering a one-stop solution for startups

"BigQuery, Looker and Firebase provided a one-stop solution that addressed all the issues faced by a growing startup," explains Gupta. "We didn't always have time to fix our platform, we're always thinking about the next group of users, and how to address challenges such as how to build revenue or obtain the next round of funding. We were able to avoid spending considerable time administering these products and, as a result, dedicated our energy to projects such as successfully securing \$60 million in Series A funding in 2021."

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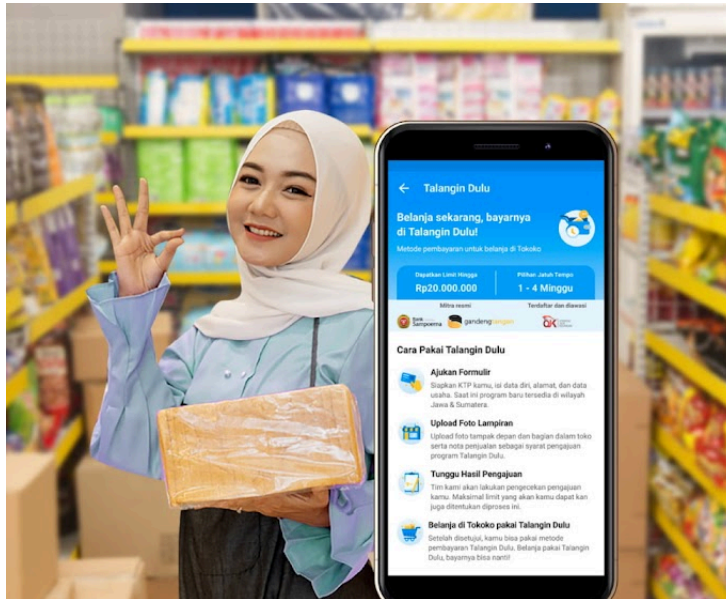
With these Google Cloud products, BukuWarung has scaled to 25 analysts supporting 300 team members overall. Gupta does not see the number of analysts increasing as the business continues to grow in size and complexity. "We have growing teams supporting multiple products, and in this scenario, most structured organizations will start throwing more data analysts, scientists and engineers at the problem," she says. "In our case, we've focused on building self-serve analytics and data literacy, so we do not envisage growing our team even as we climb to 500 team members."

Building better reports by spending less time on ad hoc requests

This self-serve and data literacy push enabled by BigQuery means BukuWarung team members are now able to scrutinize revenue and transactions without asking the data analytics team for assistance. As a result, the analytics team is spending 30% less time on ad hoc requests than previously and using that time to build better reports that support the business's drive for deeper insights.

The business is also using the cost monitoring features of BigQuery to keep unexpected cost spikes

under 15%, and optimize costs through the different pricing options provided by Google Cloud.



BukuWarung app interface for MSMEs

Tracking application performance and optimizing onboarding

BukuWarung also turned to the Firebase application development platform to monitor early stage growth and support the creation of business cases designed to secure investment in the fledgling operation. "If you're at the early stage of development when growing market share is a primary target, you need to closely track your applications," explains Gupta. "Firebase enabled us to do this and we were able to share the data captured with investors to help them understand the rich potential of our business."

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Firebase also helped the business optimize its onboarding funnel—the period from a user joining BukuWarung to making the first transaction. "Our funnel was at a constant rate of 70% and we struggled to improve this to 90%," says Gupta. "With Firebase, our product managers were able to conduct a number of experiments that included asking for one-time passwords later in the process or implementing content and know-your-customer guidance. These experiments enabled us to achieve our 90% target."

Growing payment volumes tenfold with cloud

With Google Cloud products and services operating in a multi-cloud infrastructure, BukuWarung has grown payment volumes tenfold in the last year alone, and is running successful lending experiments with fintech businesses and banks. Backed by payment, accounting and other relevant data analyzed in BigQuery, BukuWarung is well positioned to further monetize its merchant customers.

As BukuWarung continues to grow, potential use cases for other cloud products, including [BigQuery](#)

[ML](#)

(<https://cloud.google.com/bigquery-ml/docs/introduction>)

and [Dataprep by Trifacta](#)

(<https://cloud.google.com/dataprep>), emerge. "With BigQuery ML, we do see a few circumstances where our exploratory analyses could be predicted or forecast, while Dataprep has the potential to streamline and democratize derived attributes and features processing for analysts and scientists. In fact, our analysts are passionate about extracting value from both tools to grow their careers," says Gupta. One particular use case the business is considering for BigQuery ML and other Google Cloud machine learning services is developing credit scoring models to help the business determine risk in lending to individual merchants.

"We are extremely thankful for Google Cloud products and services, as well as the support we've received from sales and customer engineers to bring us to this point," says Gupta.